

NORM BROOKS

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PROFESSIONAL SUMMARY

Systems thinker with a track record of solving operational problems that conventional approaches couldn't crack. Built analytics tools that became operational infrastructure. Achieved 4x throughput improvement (8x labor productivity) by reframing a process bottleneck others treated as fixed. Maintained 99.98% inventory accuracy on \$5M of stock through forensic data analysis. Built and shipped products across systems engineering, process optimization, applied AI/LLM systems, and embedded hardware development. Repeatedly tasked with senior-level analytical and architectural work while in floor-level roles.

CORE COMPETENCIES

Distributed Systems Architecture • LLM Pipeline Development • Schema Design & Data Modeling • Embedded Systems Development (ESP32, C/C++) • Forensic Data Analysis • Process Optimization & Throughput Engineering

PROFESSIONAL EXPERIENCE

Innovairre Communications — Forest, VA

2014-2026

Operations Analyst (Production Coordinator)

Contracted by Elwood Staffing 2014-2015

High-volume direct mail production facility serving national nonprofit and commercial clients.

- Designed and built a dynamic analytics infrastructure in Google Sheets using date-partitioned data, compound-key joins, and runtime schema resolution. Provided facility management with daily production visibility across all operational areas; remains in active daily use by plant leadership.
- Selected by external operations consultant (Advatix) to extend analytics infrastructure across facilities in two states, working in lieu of contracted external labor — recognized expertise in production data and reporting systems.
- Achieved and sustained a 20% production uplift across night shift operations over a year-long period through workflow optimization and team leadership.
- Developed an SOP that quadrupled throughput on a key matching process while cutting required labor in half (creating an 8x multiplier) by reframing a manual QC step as deployable through existing automation.

H.T. Hackney Co. / Quality Foods, Inc. — Lynchburg, VA

2005 – 2013

Inventory & Production Analysis (Assistant First Shift Supervisor)

Regional distribution center for one of the largest wholesale grocery distributors in the United States, serving 20,000+ retail locations across 22 states.

- Identified inventory variance as a signal of upstream data pipeline errors rather than counting noise. Built systematic tracing methodology across receiving, PO, and billing records to eliminate error sources rather than patch outputs. Sustained 99.98% accuracy on \$5M inventory across multiple six-month cycles.
- Led a major department reorganization that consolidated a 4-person operational workflow to a single-person process by reorganizing inventory for visual identification and route optimization. 75% labor reduction while maintaining throughput and accuracy.

INDEPENDENT PROJECTS

- **Automated Database via LLM** – Prose-to-structured-data extraction pipeline with source provenance tracking.
 - [\[Link\]](#)
- **Weather Station/Climate Control Panel** – Distributed ESP32/C++ embedded system with MQTT-federated sensors and HVAC control.
 - [\[Link\]](#)
- **Pokedex Database/Calculator** – Type-matchup calculator demonstrating runtime column dispatch and schema-abstracted named ranges across 893 entries.
 - [\[Link\]](#)

EDUCATION

Associate of Business — Miller-Motte Technical College, Lynchburg, VA

Graduated Summa Cum Laude